



Artificial Intelligence Software, HW 4

Data versioning by ClearML, Labeling tools



Center for
Cognitive
Modeling



Goal of the Assignment

Short description: The task is to learn the full pipeline of preparing datasets for computer vision tasks.

- Choose either detection or segmentation problem.
- Find and analyze a public dataset.
- Collect own image data.
- Annotate images.
- Add collected and public datasets to ClearML.
- Merge datasets.
- Split dataset into train/val/test.
- Export the dataset in a format compatible with YOLO models

The final result should be a structured dataset suitable for training a neural network.



Technology Stack

The following tools should be used during the assignment.

Dataset management

- ClearML Dataset

Annotation tools

- CVAT
- Roboflow
- Supervisely

Pre-annotation

- Segment Anything Model (SAM) or others

You may use either local installations or cloud versions of these tools.



Task Selection

Tasks are listed in order of increasing complexity:

1. Object Detection
2. Image Segmentation

Annotation type depends on the selected task:

- Detection — bounding boxes
- Segmentation — masks or polygons



Step 1: Public Dataset

Find a publicly available dataset related to your selected task.

- Kaggle
- Hugging Face Datasets
- PapersWithCode
- Roboflow Universe
- Supervisely Datasets

In your report include:

- dataset name with the source link
- short description
- number of images



Step 2: Collect Your Own Images

You must collect your own dataset. The goal is to build a realistic and diverse dataset.

Requirements:

- at least 100 images
- images must be taken by you
- dataset must contain variation

Ensure diversity in:

- camera angles
- lighting conditions
- backgrounds
- scene composition



Step 3: Image Annotation

All collected images must be annotated.

Choose an annotation tool:

- CVAT
- Roboflow
- Supervisely

Annotation type depends on the selected task:

- Detection — bounding boxes
- Segmentation — masks or polygons

Ensure consistent annotation across all images.



Step 4: Create ClearML Dataset

Create a ClearML Dataset for both the public and collected datasets. And fuse them together. To sum up you should have three datasets in ClearML.



Step 5: Dataset Split

Split the dataset into three parts:

- training set
- validation set
- test set

Recommended split:

- 70% training
- 20% validation
- 10% testing

Ensure that all subsets contain representative samples.

Deserve "A" grade!

– Oleg Bulichev

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